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# Prediction of black soldier fly larval sex and morphological traits using computer vision and deep learning

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ABSTRACT

The growing interest in insect farming as a sustainable protein alternative has given rise to the commercial production of key species like the Black Soldier Fly (BSF), primarily for use in livestock, fish, and pet nutrition. Despite the heightened interest in BSF production, there is a need for increased efficiency, particularly in the context of large-scale measurement of various traits for selective breeding as well as management optimization. The unique insect production systems, coupled with the challenges posed by their small size, fragility, and metamorphic life cycle underscores the necessity for innovative approaches to streamline production.

This study explores the potential of computer vision (CV) in predicting the larval sex and morphological traits of BSF, offering a non-invasive, rapid, and automated method for trait measurement. The study explores algorithms utilizing You-Only-Look-Once (YOLOv8) in detection and segmentation, ResNet for feature extraction and classification, and regression analysis mechanisms. We assess the ability of our models to predict larval weight from images through morphometric weight prediction and CNN-regression approaches.

A notable contribution of this study is the pioneering effort to classify BSF larval sex using CV and deep learning (DL). In the analysis of larval weight prediction, a coefficient determination ($R^2$) of up to 0.80 between measured and predicted weight was achieved using the morphometric weight prediction approach, along with an $R^2$ of 0.71 through the CNN-regression approach. Additionally, the sex prediction module demonstrated an F1 score of 0.75 and a prediction accuracy of 74 %. These results underscore the feasibility of leveraging CV techniques for predicting the sex and body traits of BSF larvae, representing a significant advancement toward the automation of selective breeding in the context of insect farming.

## 1. Introduction

With the global population expected to reach 9.5 billion by 2050 [1], addressing the rising demand for protein has become imperative. Insects have emerged as a viable alternative for food and feed production [2], giving several advantages including lower greenhouse gas emissions, lower land and water requirements, and their bio-conversion capability [3]. However, the traditional practice of harvesting insects from nature, associated with seasonal dependencies [4], is considered unsustainable and poses a threat to biodiversity. Consequently, the insect farming sector for use as feed and food is steadily expanding [5], with the Black Soldier Fly (**BSF**) (*Hemertia illucens*) being one of the most farmed species [6].

Some of the challenges in the emerging insect production sector include high production costs [7] and the difficulty of guaranteeing consistent quality products across batches in large-scale production . Selective breeding has the potential to address some of these challenges: offering the opportunity to optimize quantity and quality, decrease the amount of time to harvest, and promote health and stability in insect farming [8]. Implementation of selective breeding requires, among others, large-scale measurement and recording of desirable traits, such as weight and size. Accurate measurement of traits related to growth and development not only proves pivotal for selective breeding purposes but also holds significance in the entire production management. Current approaches involve the use of manual methods such as weighing scales and rulers. However, at large scales of production, manual measurement

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can be time-consuming, and invasive, necessitating automation.

On the other hand, implementation of selective breeding schemes requires identification of selection candidates' sex at all stages where selection is implemented including during the larval stage. In addition, larval sex determination is also essential for reproductive management of the BSF colony to sustain optimal egg production through the maintenance of an appropriate ratio of males and females [9]. Bachtrog et al. [10] defined sex determination as the mechanism by which the sexual phenotype of an individual is established in a given species. In the BSF, visual sex determination is presently feasible only at the adult stage, which exhibits sexual dimorphism. This discrepancy highlights a notable gap in the current selective breeding process.

Considering the size, fragility, and short metamorphic life cycle of the BSF, a cost-effective system capable of non-invasively and accurately identifying, measuring and recording phenotypes is necessary. Advancements in the areas of computer vision (**CV**) and deep learning (**DL**) are reshaping various domains by automating processes that would otherwise be time-consuming and labor-intensive. CV and DL, as highlighted by Besson et al. [11], enable the capture, extraction, and interpretation of important features from images or videos. These techniques serve as key building blocks in intelligent automation methodologies, particularly for detection and classification tasks.

Using classical image processing and artificial neural networks, Baur et al. [12] estimated the size distribution of the *Tenebrio molitor* (mealworm) larvae using their body segments. Methods employing classical image processing are less robust in scenarios of new data and changes in imaging conditions. Also, considering the high population density of larvae and their tendency to occlude each other, the use of all the visible segments for computing the average size of the batch can be less accurate in BSF due to their deformable exoskeleton. Similarly, Hansen et al. [13] applied deep learning methods in detecting and measuring the size of BSF pupae to determine suitable larvae for breeding and consumption. The authors relied on bounding box dimensions to estimate the size of the BSF pupae, an approach that may prove ineffective for live larvae in densely crowded environments. To identify and measure the sizes of *Tenebrio molitor* larvae, Majewski et al. [14] used Mask R-CNN for instance segmentation and developed a larvae phenotyping model to infer the length and mass of larvae. Baur et al. [12] and Majewski et al. [14] measure the size and track growth in *Tenebrio molitor* larvae, utilizing the body segments which is possible due to the rigid larval exoskeleton. However, in BSF larvae, the deformable body structure in response to stimuli or activity makes it challenging to rely on the size of body segments for morphological trait prediction. Also, [15] based on the larval body width to develop the length and volume prediction models of the *Tenebrio molitor* larva using a regression Convolutional Neural Network (**CNN**) and this was later modified in [16] to use YOLOv8 for the end-to-end monitoring of *Tenebrio molitor* growing. Considering the larval shape, Smiley and Wisdom [17] proposed methods to estimate the mass of insect larvae based on the volume-mass relationship, on the assumption that the larva segments are cylindrical. Similarly, Rogers et al. [18] and Sample et al. [19], indicate that there exists a relatively high correlation between insect biomass and body length, making the length a considerable feature for predicting body weight.

In this study, we utilize the YOLOv8 [20,21] model for detection and segmentation. The aim is to investigate the potential of DL and CV approaches for the measurement and prediction of individual BSF larvae morphological traits such as weight and length from digital images. The study also explores the potential of deep learning in predicting the sex of the BSF at the larva stage.

Two approaches are explored to predict larval weight. In the first one, we utilize a DL segmentation model and image analysis techniques to extract size features, which is referred to as the "morphometric prediction". In the second approach, we use a detection model to preprocess images as input to a ResNet [22] pre-trained model for individual larva weight prediction which is referred to as the "CNN-regression approach". Similarly, for BSF larval sex determination, a classification model based on the ResNet is implemented, exploring the high-level feature extraction capabilities of the pre-trained model.

## 2. Methods and materials

The general overview of steps involved in the data acquisition, model training and testing, are presented in Fig. 1.

### 2.1. BSF samples

For this experiment, a BSF colony originally obtained from Enorm Bio-factory (Flemming, Denmark) was maintained at the insect laboratory at the Institute of Biology of Aarhus University. The colony was maintained in a climate-controlled room at 14:10 h light:dark cycle, a temperature ranging between 25 and 29 °C, and relative humidity (RH) of approximately 60 % [23]. The BSF colony was fed on a chicken feed (CF Pacostar, DLG, Fredericia, Denmark) diet. Data collection (imaging and manual measurement) was undertaken on larvae samples from 16 batches of the colony sampled between June and October 2023. To increase the representation of different sizes, the age of sampled larvae ranged between 11 and 14 days after hatching.

### 2.2. Image data acquisition system

An imaging box, housing a camera station, was purpose-built to maintain consistent illumination conditions. The camera station comprised of a Raspberry Pi 3 module equipped with a Raspberry Pi HQ 12.3 MP camera combined with a 3MP 8–50 mm industrial zoom lens. Inside the box, a cold white 5 V LED strip is installed to provide diffuse light. The system is connected to a display screen, and a push-button linked to the Raspberry Pi to afford additional control during image capture. The camera is positioned 70 mm above the larva imaging area, a height determined empirically to ensure that the larvae captured in the images were of an appropriate size.

### 2.3. Dataset and reference data collection

In pursuit of a robust detection and segmentation model, we captured 321 larvae images with over 5000 larvae instances to train and validate the YOLOv8 model. Additionally, to establish reliable ground truth data for training and validating the prediction models, reference data was collected manually, including larval body weight, length, and sex. Larval body weight was measured using a digital microscale with ±1 mg precision while a digital caliper was used for length measurement. Throughout data collection, each larva was allocated a unique identifier and was manually measured for body length and weight before being put into the imaging station for image acquisition. Subsequently, the individual larva was carefully transferred to an appropriately labeled vial to facilitate tracking until its emergence as an adult, at which point its sex was recorded. In total, images and reference data on body weight, length, width, and larval sex were available for 1600 individual larvae.

### 2.4. Image analysis

As illustrated in Fig. 1, two different approaches were used in predicting the weight of the BSF larva which involved different analysis steps respectively.

#### 2.4.1. Detection and segmentation

Detection enables locating objects within an image while segmentation locates pixels belonging to a particular class or instance in an image [24,25]. The YOLOv8 segmentation model selected for this study belongs to the growing family of the YOLO models [26]. YOLO reframes object detection as a single regression problem, straight from image

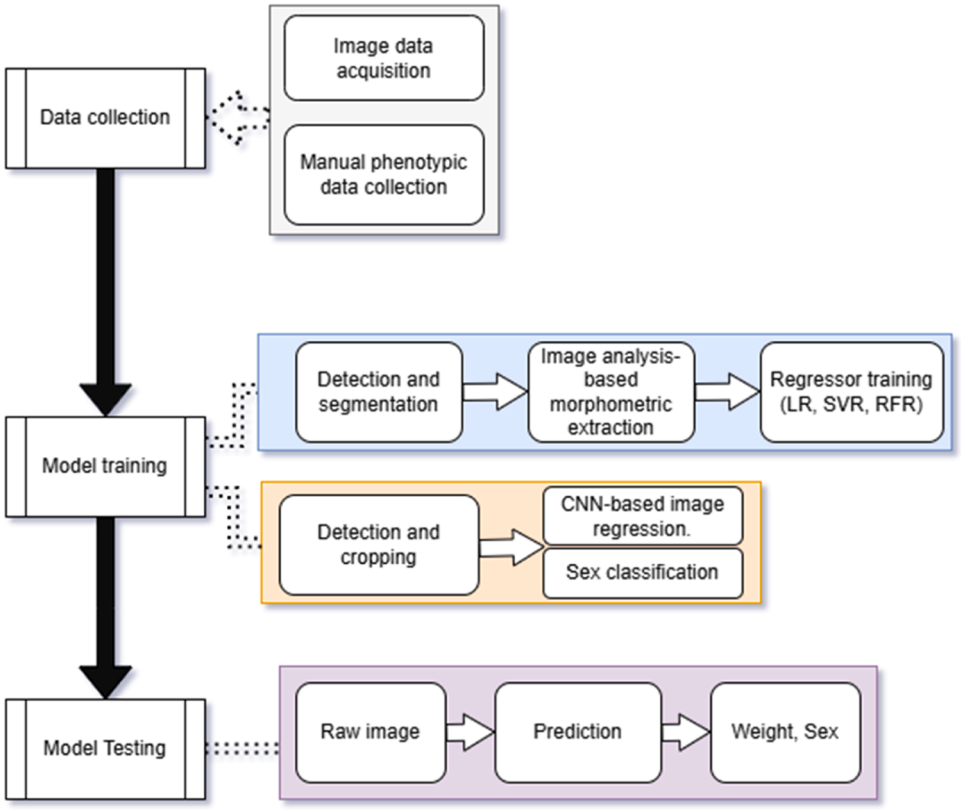



**Fig. 1.** An overview of the methods and approaches used in the weight and sex prediction. This highlights the data collection, model training and the model testing processes.

pixels to bounding box coordinates and class probabilities. The YOLOv8 segmentation model extends the YOLOv8 object detection framework by adding the capability to generate masks for each detected object.

To prepare images for training and validating the YOLOv8 segmentation model, image annotation was done using *Labelme* [27] an open-source software. A dataset (**DS1**) with 321 images was used for training and testing the detection and segmentation model. The 4xNVIDIA TITAN X (Pascal) (4 × 12 GByte memory) GPU was used to train the model for 100 epochs using the YOLOv8m-Seg model pre-trained on the COCO image dataset [28]. The original images were resized to 640 × 640 for training. The details of the DS1 dataset used in training and validation are indicated in Table 1.

The trained segmentation model was then applied to the single larva images dataset (**DS2**) for feature extraction, and the extracted features were used for the morphometric weight prediction.

### 2.4.2. Detection and cropping

To generate a dataset (**DS3**) for predicting larval weight and sex through CNN-regression and classification respectively, the original images in DS2 of size 4056 × 3040 were preprocessed using the YOLOv8 detection model as highlighted in Fig. 1 and illustrated in Fig. 3. The bounding boxes from larva detections were cropped out, generating new images with varying sizes. In total, a dataset of 1600 images was generated as shown in Table 1 for the CNN-regression (weight prediction) and classification (sex determination) tasks.

**Table 1**
Number of items in the training and test datasets.

| Dataset type | Yolov8 segmentation model (**DS1**) (No. of images) | Weight Prediction (No. of larvae) | | Sex determination (No. of larvae) |
|---|---|---|---|---|
| | | Morphometric Prediction (**DS2**) | CNN-regression prediction (**DS3**) | |
| Train | 224 | 1280 | 1280 | 560 |
| Test | 97 | 320 | 320 | 140 |

## 2.5. Larval weight prediction

Two approaches were implemented for predicting the live weight of BSF larvae from images: the morphometry and the CNN-regression methods.

### 2.5.1. Morphometric weight prediction

Morphometric weight prediction involved extracting key morphological features from segmented larval images to assess their relationship with larval weight. The process consisted of image segmentation, feature extraction, and regression analysis to develop predictive models.

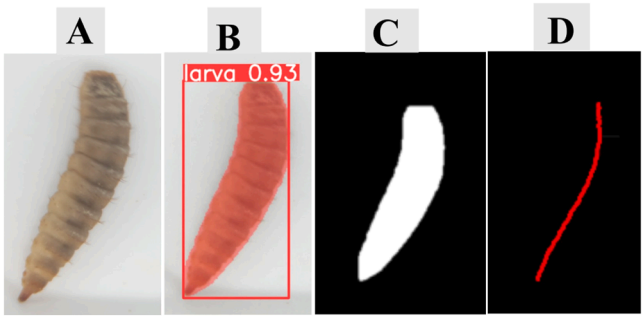



**Fig. 2.** Feature extraction transition of the images. (A): Original BSF larva. (B): Segmented larva. (C): Binary mask of the larva, where the area is extracted. (D): skeletonization results from the mask to measure the length of the larva.

Using the original larval image (Fig. 2A), segmentation was performed with YOLOv8 (Fig. 2B) to generate a binary mask (Fig. 2C). Morphometric features such as area, contour length (perimeter), length, and width were extracted and evaluated for their effectiveness in predicting larval weight.

Extracted feature include:

- Contour Area: Extracted using OpenCV, representing the total pixel area of the segmented larva.
- Length Measurement: Computed using the skeletonization algorithm [29] (Fig. 2D). This process iteratively removes contour pixels to create a centerline, which is used to estimate larval length. Since skeletonization can introduce unwanted branches that affect length estimation, the Fil-finder library [30] was applied to prune the skeleton and refine the measurement.
- Width Calculation: The larval width is determined using the distance transform method, where the maximum distance from the skeleton represents half of the total width. This value is then multiplied by 2 to obtain the full width. This approach assumes that the skeleton is centrally located within the larva and accounts for its slight curvature.

All extracted features were initially measured in pixels and then converted to millimeters (mm) using a calibration factor of 28 pixels per mm, derived from a standard-sized reference object in the images.

To investigate the predictive value of different morphometric features, three feature combinations were evaluated:

- Area and length
- Area and width
- Area, length, and width combined

Feature data was extracted from 1600 larvae (Table 2) and used to train regression models for weight prediction. The dataset was split into 80 % training and 20 % testing to ensure reliable evaluation.

Three regression algorithms were implemented:

- Support vector regressor (SVR) [31]
- Linear regression (LR)
- Random forest regressor (RFR) [32]

To optimize model performance and determine the best hyperparameters for SVR and RFR, a five-fold grid search cross-validation was applied within the training dataset. The selected hyperparameters were:

- SVR: $C = 100$, $\epsilon$=0.01, kernel = radial basis function (RBF).
- RFR: max depth = 10, min samples leaf = 2, min samples split = 10, Once the optimal hyperparameters were identified, the models were retrained on the entire training set, and their final performance was evaluated using the test set.

#### 2.5.3. CNN regression-based prediction

In this approach, larval weight was predicted directly from images using a deep learning-based regression model. Unlike the morphometric method, which relies on manually extracted traits, this approach leverages convolutional neural networks (CNNs) to learn complex visual patterns associated with weight variations.

To build the dataset, individual larval images were detected and cropped using the trained YOLOv8 detection model, generating a collection of images with varying sizes. The goal was to train a deep learning model that could infer weight based on the visual characteristics of each larva.

The CNN-regression model consisted of two main components: a feature extractor and a regression module. Several pre-trained CNN architectures, including VGG16, MobileNet, and ResNet, were evaluated for feature extraction. The final model adopted a ResNet101 backbone [22], pre-trained on the ImageNet dataset [33], to extract high-level features from the images.

To enhance prediction accuracy, a hybrid approach was used:

1. Feature extraction: The ResNet101 model extracted deep image features while preserving spatial and structural information.
2. Regression analysis: These combined features were fed into a support vector regressor (SVR), which was fine-tuned using grid search cross-validation to optimize hyperparameters (Table 2).

Since ResNet101 requires fixed-size inputs, all images were resized to 224 × 224 pixels, while their original dimensions were stored and later incorporated into the regression process. The entire process is illustrated in Fig. 3(A), and dataset details are provided in Table 1.

This end-to-end approach enabled the model to automatically learn relationships between model-defined features from images and larval weight, reducing reliance on manually extracted features while improving predictive performance.

### 2.6. Larval sex prediction

The sex prediction model was developed using the generated image dataset from the detection and cropping model. Two approaches were explored in this regard, which involved (1) an "Image-only" CNN model that used the larvae images as the only inputs to perform sex classification as illustrated in Fig. 3**(B)**, and (2) a fusion model combining CNN model and a Multi-Layer Perceptron (**MLP**) model as illustrated in Fig. 3 **(C)**. A fusion model was developed in an effort to enhance sex prediction accuracy by leveraging morphological traits, combining size information (weight and length) with image data. Training was performed by optimizing the different training hyperparameters such as the learning rate, batch size, and regularization parameters, and employing techniques such as stratified sampling and weighted random sampling due to the imbalances in the dataset. The original dataset consisted of 1200 images and was later reduced to 309 male and 386 female BSF larvae after different training trials to balance the dataset. The images were preprocessed by padding based on the aspect ratio and resized to 224 × 224 as the standard input in ResNet models. The training/test split is 80/20 as illustrated in Table 1. In the first modality, the ResNet18 pretrained model was fine-tuned and a ReLU activation layer and a dropout layer of 0.2 were added before the final classification layer. The binary cross entropy loss criterion with stochastic gradient descent (SGD) optimizer, with a learning rate = 10–4 and a momentum = 0.9 was adopted for model training. In the second modality, an ensemble model combining the ResNet18 model output with the MLP model output was trained. Both models output vectors of 256 which were combined to a binary output. Similar to the first modality, adjustments to the model parameters were undertaken during model training to find the most suitable model for sex prediction. The model was trained for 30 epochs, beyond this the model had started overfitting to the training data.

**Table 2**
Weight prediction results from the area and length features with regression analysis.

| | Area and width | | Area and length | | Area, length and width | |
|---|---|---|---|---|---|---|
| Model | RMSE (g) | $R^2$ | RMSE (g) | $R^2$ | RMSE (g) | $R^2$ |
| LR | 0.25 | 0.61 | 0.019 | 0.77 | **0.018** | **0.80** |
| SVR | 0.25 | 0.60 | 0.019 | 0.77 | 0.019 | 0.79 |
| RFR | 0.25 | 0.59 | 0.020 | 0.76 | 0.019 | 0.77 |

RMSE= Root Mean Squared Error; LR = Linear Regression; SVR= Support Vector Regressor; RFR = Random Forest Regressor, $R^2$= coefficient of determination.

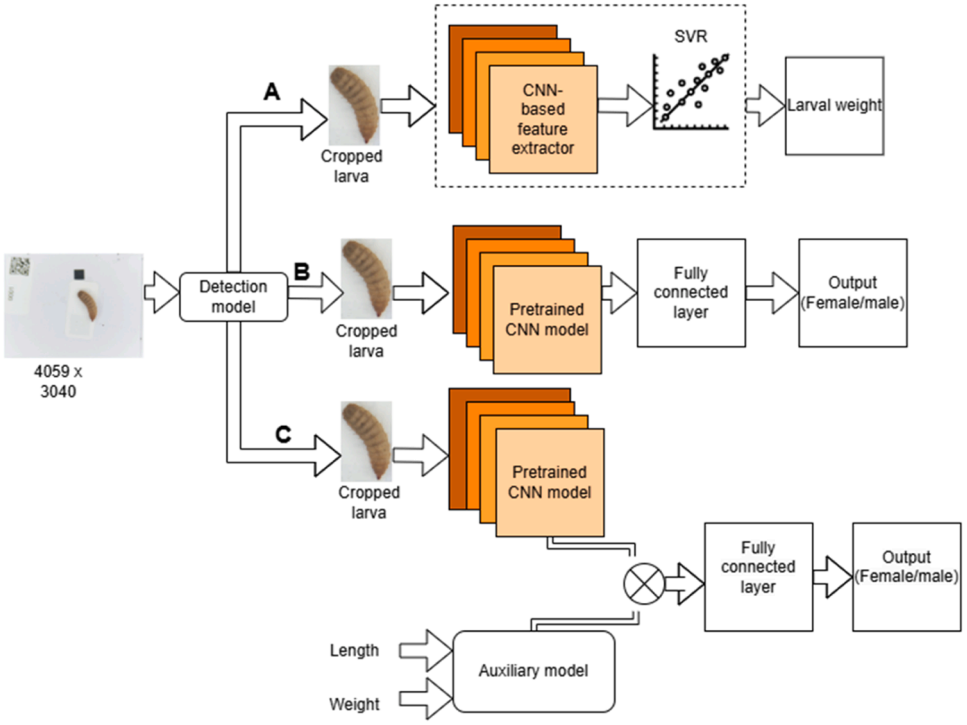



**Fig. 3.** CNN-based weight and sex prediction approaches, (A) Process flow for the weight prediction using the CNN-regression method. (B) Image only CNN-based sex prediction. (C) Fusion model sex prediction approach.

### 2.7. Evaluation

To assess the performance of the prediction models, evaluation metrics were selected based on the tasks at hand. The evaluation was done using test datasets that had not been seen during the training of the models. The evaluation focused on three key aspects: (1) the accuracy of larval detection and segmentation, (2) the performance of larval weight prediction using the morphometric as well as the CNN-regression approaches, and (3) the accuracy of image-based larval sex classification models.

#### 2.7.1. Larva detection and segmentation model evaluation

To evaluate the performance of the YOLOv8 model in detecting and segmenting larvae from images, average precision **(AP)** was used as the primary metric. AP quantifies the model's segmentation accuracy by computing the area under the precision-recall curve, considering different confidence thresholds. Correct predictions are determined based on the Intersection-over-Union (IoU) threshold, which is set to 0.5 to ensure accurate segmentation results.

$$Seg(mAP) = \sum_{i=1}^{c} \frac{AP(i)}{c},$$

Where c denotes number of classes.

Since this task involves only one class (larvae), the mean average precision (mAP) simplifies to the average precision (AP) for that class,

#### 2.7.2. Weight prediction

The performance of the weight prediction models was evaluated using two key metrics:

1. Coefficient of determination ($R^2$) – a measure of the proportion of variance in actual weight that is captured by the model:

$$R^2 = 1 - \frac{\sum (y_i - \widetilde{y_i})^2}{\sum (y_i - \bar{y})^2}$$

2. Root mean squared error (RMSE) – a measure of the average magnitude of prediction errors, providing an indication of how closely the predicted weights align with actual values:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \widetilde{y_i})^2}$$

Where, $n$ = number of records (larvae) in the independent test set, $y_i$ = actual (measured) weight of $i^{th}$ larva in the test set, $\widetilde{y_i}$ = predicted weight of $i^{th}$ larva in the test set, $\bar{y}$ = mean of the actual weights for larvae in the test set.

Both morphometric prediction and the CNN-regression approach were assessed using these metrics to assess their predictive performance for larval weight. Performance evaluation was conducted using a test dataset, which was not seen during model training, ensuring a fair and unbiased assessment of model generalization.

#### 2.7.3. Sex classification

The performance of the sex classification models was evaluated using three key metrics:

1. Accuracy – a measure of the proportion of correctly classified larvae:

$$Accuracy = \frac{\text{All correct predictions}}{\text{All predictions}} = \frac{\text{TP} + TN}{TP + TN + FP + FN} \#$$

where TP (true positives) and TN (true negatives) represent correctly classified male and female larvae, respectively, while FP (false positives) and FN (false negatives) indicate misclassified instances.

2. Precision and recall – Provide deeper insights into classification performance:

- Precision: The proportion of correctly predicted positive instances (e.g., males or females) among all predicted positives.
- Recall: The proportion of correctly predicted positive instances among all actual positives.

3. F1 score – A harmonic mean of precision and recall, balancing the trade-off between false positives and false negatives:

$$F1 = 2\ x\ \left(\frac{Precision\ x\ Recall}{Precision + Recall}\right)$$

A confusion matrix was used to visualize classification performance by comparing actual versus predicted sex labels. Performance evaluation was conducted using an independent test dataset, ensuring an unbiased assessment of the model's ability to generalize to unseen data.

## 3. Results

### 3.1. Detection and segmentation

The YOLOv8 model attained a segmentation AP50 of 0.93. This means that the model can accurately locate and segment the larvae from the images. Fig. 4 illustrates the detection and segmentation process. In the first case, the model segments a single larva with both a mask and a bounding box. In the second case, the larva is detected using only a mask.

### 3.2. Weight prediction

A summary of the results from the weight prediction models is presented in Table 2 indicating the various feature combinations and the regression models explored. As presented in Fig. 5, the LR model, utilizing the area, length and width features demonstrated a predictive performance for weight, achieving a RMSE of 0.018 with an $R^2$ value of 0.80. The SVR yielded a RMSE of 0.019 and $R^2$ of 0.79 while the RFR model exhibited a RMSE of 0.77 with $R^2$ of 0.019 with the associated RMSE. Results from other feature combinations such as "length and area", and "length and width" are shown in Table 2.

In contrast, the performance of the CNN-regression approach for weight prediction is presented in Fig. 6. The model achieved $R^2$ of 0.71 and RMSE of 0.023, indicating an average performance in the prediction of larval weight.

### 3.3. Larval sex classification

The sex classification results are presented in the confusion matrices in Fig. 7, illustrating the classification performance of the selected approaches. These metrices compare the "True" male and female class against the "predicted" classifications. In the "image-only" approach (Fig. 7**A**), the model achieved a larval sex classification accuracy of 74 % with the F1 score of 0.75, indicating that it correctly classified 74 % of both male and female larvae from images. In the second approach, which employs a fusion model, larval sex classification accuracy of 69 % was achieved (Fig. 7**B**).

Note that the total test sample in the image-only approach is 139 and the fusion model approach is 136. The difference is attributed to variations in data loading.




**Fig. 4.** Detection and segmentation of larvae from an image using the YOLOv8 model. The top row indicates the raw unprocessed images while the bottom row depicts larva segmentation masks (with a bounding box and a confidence score) and without the bounding box.

## 4. Discussions

### 4.1. Detection and segmentation

Detection and segmentation are pivotal for extracting larval pixels from images. Recent studies such as Baur et al. [12] utilized classical image processing techniques to segment mealworm larvae, whereas Majewski et al. [14] and Majewski et al. [16] employed the Mask R-CNN model for similar tasks. Our study pioneers the use of the YOLOv8 model for the detection and segmentation of BSF larvae, marking a significant advancement in this field. While previous research primarily focused on the mealworm larvae, with notable exceptions like Hansen et al. [13] achieving BSF larva detection rates of up to 87 % mAP using SSD-MobilenetV2, our work expands this scope considerably. The YOLOv8 model, built to strike a balance between accuracy and speed, operates as a single-stage model capable of real-time detection and segmentation, making it well-suitable for applications in insect production contexts. Our trained model demonstrates precise larval instance detection achieving an AP50 of 0.93, indicating a high accuracy in detecting and localizing larvae. This accuracy extends to effectively segmenting detected larvae accurately outlining the larva pixels against white backgrounds. Further refinement and evaluation of the YOLOv8-segmentation model on larvae in dense scenes will contribute to its robustness and applicability in actual BSF production settings which will aid in measuring group means of the body traits.

### 4.2. Larval weight prediction

The application of DL, CV and machine learning for weight prediction has gained significant interest across various domains including livestock production [34,35], fish [36,37], seed quality assessment [38] and in entomology [39,40]. In BSF larvae, weight is largely influenced by genetics, age, substrate type, or other environmental factors. Predicting larval weight from images typically relies on morphometric traits such as area, length, width, or volume. Regression analysis based on morphometric traits has been successfully applied in the prediction of weight in different fields [41–43]. To demonstrate the effectiveness of image-based weight prediction models, Majewski et al. [14] utilized the length and volume to calculate the mealworm larval mass. In our study, the results from the morphometric approach demonstrate that the LR model outperforms both SVR and RFR in terms of RMSE and $R^2$ across all variable combinations, indicating its effectiveness in capturing the relationship between variables. An RMSE of 0.018 suggests that, on average, the model's predictions deviate from the actual weights by 0.018 *g* Given that most true weights range between 0.1 g and 0.3 g, this error accounts for approximately 9 % of the total range. This suggests that the model demonstrates a relatively high accuracy in predicting larval weight. Among the feature combinations explored, including length alongside area consistently enhanced model performance, while width had a lesser impact. The distance transform method used to estimate larval width is sensitive to noise and computationally intensive, which can lead to inaccuracies and lower response times. Thus, length and area are identified as key variables for predicting larval weight, as notable model performance is observed when these features are used in combination.

However, accurately measuring the length of BSF larvae is challenging due to their tendency to retract in response to tactile stimuli

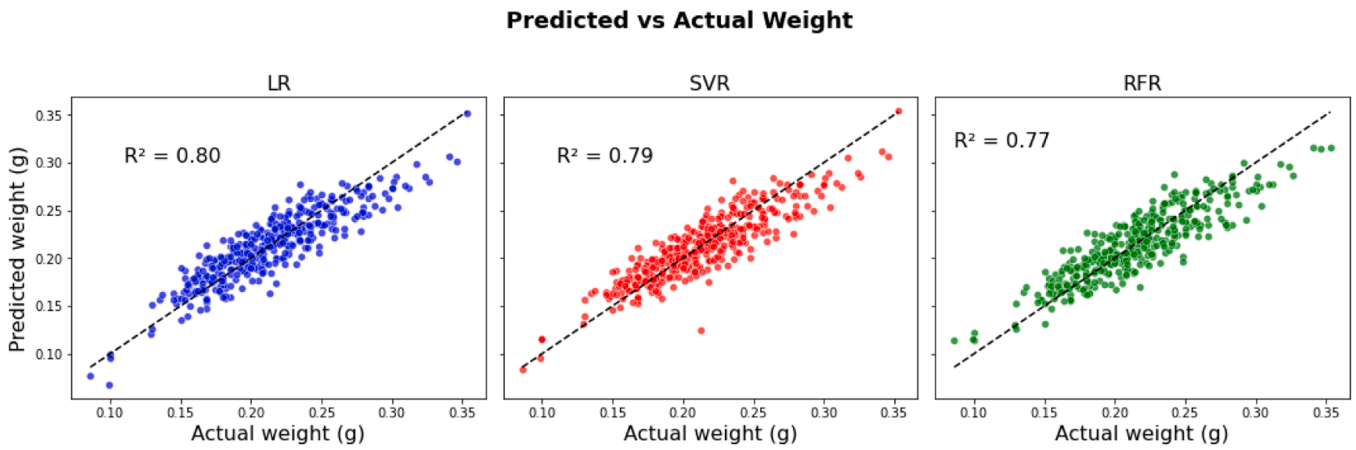



**Fig. 5.** Plot of the predicted and actual weight results for the area, width and length feature combination. (A) the linear regression plot. (B) Support vector regressor. (C) random forest regressor.

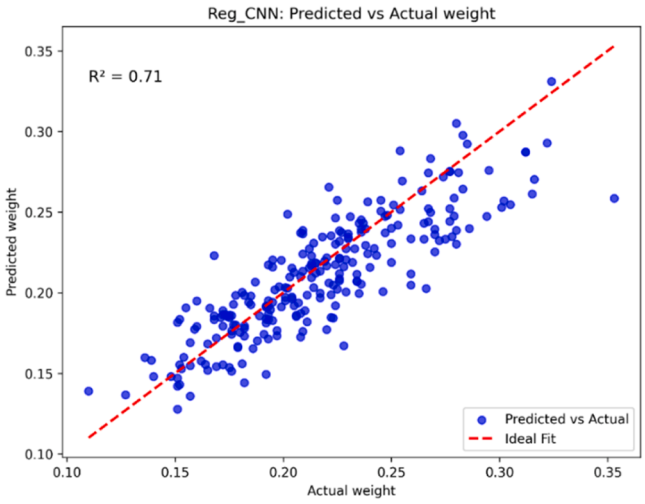



**Fig. 6.** A plot of the actual vs the predicted weight using the CNN regression approach.

during manual measurements and imaging, causing a random discrepancy between manually measured length and digital image-based length. Furthermore, the skeletonization algorithm's sensitivity to image noise may introduce errors in length estimation, particularly with small objects like BSF larvae. Ultimately, the choice of features and models depends on the specific traits of interest. The morphometric approach is well-suited for large-scale production because of its flexibility, enabling the study of various traits important for selective breeding. Additionally, required size measurements can be adjusted according to the pixel-to-mm conversion factor, making this approach adaptable and scalable.

[15] employed a regression CNN and knowledge transfer techniques to estimate mealworm larval size distribution based on segment width. In our study, the CNN_ regression model for weight prediction demonstrated a strong correlation between the actual and predicted weights, as evidenced by the clustering of scatter plot points around the line of best fit. This suggests a positive relationship between true and predicted weights, highlighting the potential of this approach for non-invasive weight prediction.

A few outliers were observed, which may result from measurement errors, data anomalies, or the model's limitations in capturing more complex relationships. With high-quality images, this approach could be extended to various applications, such as predicting larval quality based on color pigmentation or body segment depth. However, this requires consistent imaging conditions and retraining for images captured at different scales, posing a challenge for large-scale production scenarios.

### 4.2. *Larval sex classification*

This study pioneers the exploration of predicting larval sex from

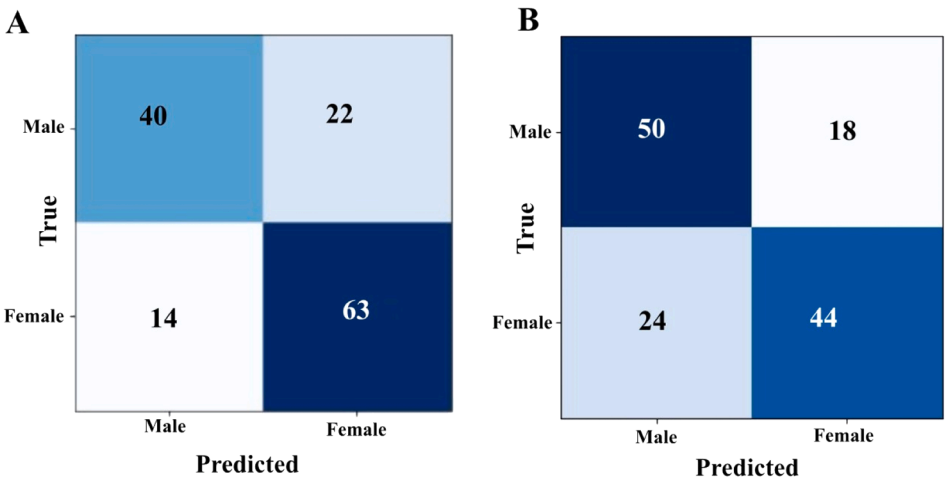



**Fig. 7.** Confusion matrices of BSF larvae sex prediction. (A) with the image-only model and (B) based on the fusion sex prediction model.

single 2D images using computer vision (CV) and deep learning (DL), representing a novel approach to developing low-cost, fast, and accurate sex prediction models. Our sex prediction results depict great potential for using high-quality images in larval sex determination. With an observed accuracy of 74 %, it is noteworthy to mention that by chance, the prediction would be 50 %. This accuracy indicates that approximately 7 and a half out of 10 predictions made by the model are correct. Moreover, the model demonstrates a good balance between precision and recall, although it may misclassify some instances from the classes. The model is better at correctly predicting the 'True' class, even if overall accuracy is lower, and it outperforms a human random guess. The method using only images in sex determination outperforms the combined-model method, illustrating the CNNs' learning abilities from high-level image features, implying that images hold enough information for sex determination. Currently, sex determination at larval and pupal stages is challenging due to the apparent absence of visible secondary sexual characteristics. This inefficiency highlights the necessity for a more dependable, cost-effective, and robust solution for sex determination in BSF larvae.

Certain features like depth of larva segments, variations in length and width, and colorations could be well learned with deep learning models allowing automated identification of sex at the larval stage. The findings from this study present a promising low-cost adaptable solution to the sex determination problem. This could revolutionize the process by providing farmers and researchers with a low-cost and reliable method of sex determination thereby positively impacting various aspects of BSF rearing and research.

Although our study offers initial insights into the potential of using CV and DL for predicting BSF larval sex, it falls short in thoroughly examining the specific parameters and features employed by the model in making predictions. Thus, it is crucial to interpret these findings with caution, recognizing the need for further investigation into the key features enabling sex determination at the larval stage of BSF. Additional studies are warranted to delve deeper into this aspect and elucidate the underlying mechanisms driving CV-based sex prediction in BSF larvae.

### *4.3. Practical implications and way forward*

Our study presents an initial proof of concept on the possibilities of computer vision-based live weight prediction and sex classification in BSF larvae. While the assumptions and conditions under which this proof of concept was conducted are far from actual production scenarios, it provides a foundational overview of the potential for automating phenotypic trait measurement in insect farming. The findings demonstrate that CV and DL can offer non-invasive, rapid, and accurate methods for predicting larval weight and determining sex, which are critical for optimizing breeding programs and improving production efficiency. However, several limitations and challenges must be addressed to bridge the gap between laboratory-scale experiments and real-world applications.

One key limitation is the focus on individual larvae weight prediction, which does not fully align with actual breeding scenarios where larvae are often densely packed in substrates. This raises questions about the scalability of the proposed methods in large-scale production environments. Additionally, the image processing techniques used for measuring larval length and width are computationally intensive, which could hinder real-time applicability in high-throughput systems. Another challenge lies in the "black box" nature of the sex classification model, as it remains unclear which specific features of the larvae are being used for classification. This limits the interpretability and generalizability of the model, particularly when applied to larvae reared under different conditions. Furthermore, the dataset used in this study is skewed, with most larvae weighing between 0.1 and 0.3 g, potentially restricting the model's ability to generalize to larger larvae or other insect species.

Despite these challenges, the study opens up several promising avenues for future research and practical applications. For instance, future work could explore the detection of larvae mixed within substrates in actual rearing trays, addressing the issue of occlusions and overlapping larvae that are common in real-world farming conditions. Comparing the performance of two-stage deep learning models, such as Mask R-CNN, with YOLOv8 for detecting multiple larvae instances could provide valuable insights into the trade-offs between accuracy and computational efficiency. Improving size measurement methods by employing key-point detection techniques could also enhance the accuracy of trait extraction, particularly in dense conditions where larvae are not fully visible. Refining the sex prediction models by testing them on larvae reared in different environments and with varied feed would improve their robustness and generalizability. Additionally, expanding the dataset to include a wider range of larval sizes and conditions would further enhance the accuracy and reliability of both weight and sex prediction models.

Ultimately, the integration of these methods into real-world insect farming systems could revolutionize the way phenotypic traits are measured and monitored, enabling farmers to make data-driven decisions that improve yield, sustainability, and overall production efficiency. By addressing the current limitations and building on the initial findings of this study, future research can pave the way for more efficient and sustainable insect farming practices, contributing to the global demand for alternative protein sources.

## 5. Conclusion

This proof-of-concept study explores the potential application of computer vision (CV) and deep learning (DL) for the non-invasive prediction of larval sex and live weight in the black soldier fly (BSF). By integrating YOLOv8 for precise detection and segmentation, ResNet architectures for feature extraction, and regression models, we demonstrate promising results in predicting larval weight ($R^2$ up to 0.80 using morphometric methods and 0.71 via CNN- regression) and sex classification (74 % accuracy with an F1 score of 0.75). These findings demonstrate the feasibility of using CV/DL to replace traditional, labor-intensive phenotyping methods in insect farming.

Successful development of CV based larval sex and weight prediction approaches address key challenges in BSF production, such as scalable trait measurement for selective breeding and early sex determination, which are essential for improving reproductive management and yield optimization. While this study provides a foundational proof of concept in controlled conditions, future research should aim to enhance model generalizability in dense, real-world production environments, improve computational efficiency for high-throughput systems, and explore the biological features driving sex prediction. Bridging these gaps will enable CV-driven automation to advance sustainable insect farming, supporting global food security and circular bioeconomy objectives.

## CRediT authorship contribution statement

**Sarah Nawoya:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Quentin Geissmann:** Writing – review & editing, Supervision, Software, Methodology. **Henrik Karstoft:** Writing – review & editing, Supervision, Methodology. **Kim Bjerge:** Writing – review & editing, Supervision, Methodology. **Roseline Akol:** Writing – review & editing, Supervision. **Andrew Katumba:** Writing – review & editing, Supervision. **Cosmas Mwikirize:** Supervision. **Grum Gebreyesus:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data is available online at DOI: 10.5281/zenodo.13359376.

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